



Department of Artificial Intelligence and Machine Learning
Data Modeling and Visualization Laboratory (AIL48)

I.A. Marks: 50

Exam Hours: 03

Credits: 0:0:1

Exam Marks: 50

Part A

1. Read N number from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable message.
2. Write a python program to read 4 subject marks and display grades. (91-100: S Grade, 81-90: A Grade, 71-80: B Grade, 61-70: C Grade, 51-60: D Grade, 41-50: E Grade, 00-40: F Grade)
3. Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with suitable message
4. You are provided with a number D containing only digits 0's and 1's. Your aim is to convert this number to have all the digits same. For that, you will change exactly one digit i.e. from 0 to 1 or from 1 to 0. If it is possible to make all digits equal (either all 0's or all 1's) by flipping exactly 1 digit then output "YES", else print "NO" (quotes for clarity). Input Format: The first line of the input contains the number D made of only digits 1's and 0's. Output: Print 'YES' or 'NO' depending on whether it's possible to make it all 0s or 1s or not. Example-1: Input: 101 Output: YES, Example-2: Input: 11 Output: NO.
5. Write a program in python for the following variable ids refers to the list [4353, 2314, 2956, 3382, 9362, 3900]. Using list methods, do the following:
 - i. Remove 3382 from the list.
 - ii. Get the index of 9362
 - iii. Insert 4499 in the list after 9362
 - iv. Extend the list by adding [5566, 1830] to it
 - v. Reverse the list
 - vi. Sort the list
6. Given an Input File Which Contains a List of Names and Phone Numbers Separated by Spaces in the Following Format:

- i. Alex 80-23425525
- ii. Emily 322-56775342
- iii. Grace 20-24564555
- iv. Anna 194-49611659

Phone Number Contains a 3- or 2-Digit Area Code and a Hyphen Followed by an 8-Digit Number. Find All Names Having Phone Numbers with a 3-Digit Area Code Using Regular Expressions.

7. Write a Python program that ask the user to enter the name of a text file. Display the specified number of lines from the input file, calculate and display the frequency of the specified word in the file.
8. Write a Python Program to Check the Validity of a Password Given by User. The Password Should Satisfy the Following Criteria:
 - Contain at least 1 letter between a and z
 - Contain at least 1 number between 0 and 9
 - Contain at least 1 letter between A and Z
 - Contain at least 1 character from \$, #, @
 - Minimum length of password: 6
 - Maximum length of password: 12
9. Write a program to make a new string with all the consonants eliminated from the string read from the user. For example:

Input: Hello, have a good day
Output: Hll, hv gd dy
10. Write a Python program that processes a text file named 'content.txt'. Perform the following task
 - a. Extract and Sum Numbers
 - b. Extract Dates and display the same

Part B

1. Develop a python program to create a class called as Employee with attributes like EmpID, EmpName, EmpDesignation, Experience and Age. Write the methods for addEmployee(), displayDetails() and calculateSalary(float basic). The salary of an employee should be calculated based on the following criteria:
 - If the employee is below 30 years with 5+ experience, then the final salary 1.5 * basic.

- If the employee is below 40 years with 5+ experience, then the final salary 1.75* basic.
 - If the employee is below 40 years with 10+ experience, then the final salary 2 * basic.
 - If the employee is below 50 years with 20+ experience, then the final salary 2.25 * basic.
 - If the employee is below 50 years with 25+ experience, then the final salary 2.5 * basic.
 - If the employee is below 58 years with 30+ experience, then the final salary 3 * basic.
2. Develop a python program to create a class called as "FourthSem" with attributes like RollNum[20], Test1Marks[20], Test2Marks[20] and Test3Marks[20]. Calculate class average for each of the three tests for a class of 20 students. The program should also calculate the average of each of the student scores in those three tests and display. Display the top 5 and last 5 scores of the class for every test.
3. Consider the following marks scored by students in various subjects:

Student ID	Subject	Marks
1	English	78
2	Mathematics	92
3	Physics	36
4	Chemistry	64
5	Biology	89

Represent the above data in a 1-dimensional array where each element corresponds to the marks in the order given. Calculate the sum of all the marks. Additionally, award extra marks to each subject as follows:

- English: +2
- Mathematics: +2
- Physics: +5
- Chemistry: +10
- Biology: +2

Write a Python program to perform these operations.

4. The engineers at XYZ Custom Cars want to analyse the horsepower data of their latest car models. Given a list of 5 horsepower values, calculate and provide the mean and median of these values using NumPy libraries.

horsepower = [130, 165, 150, 150, 140]

5. Perform a Student Performance Analysis using given data on student grades in different subjects. The analysis should include the following visualizations:
- Compare the distribution of grades across different subjects using a box plot.
 - Show the relationship between grades in two specific subjects using a scatter plot.
 - Display the distribution of overall grades using a histogram.
6. Given data on website traffic over a period of time, create visualizations such as: Line plot showing the daily number of visitors. Bar plot comparing the number of visitors on weekdays versus weekends. Pie chart showing the percentage of traffic from different sources (direct, search engines, social media, etc.).
7. Write a python program to load the CSV file into a Pandas Data Frame. Perform the following task:
1. Check for missing values and drop the corresponding rows.
 2. Find the district with the highest annual rainfall.
 3. Display the top 5 states with the highest annual rainfall.
 4. Drop the specified columns ('Jan-Feb', 'Mar-May', 'Jun-Sep', 'Oct-Dec').
 5. Create a pivot table to display state-wise mean rainfall for all the months.
 6. Count the number of districts in each state.
 7. For each state, find the district that gets the highest rainfall in May and display the recorded rainfall.
8. Implementing Web Scraping in Python with BeautifulSoup

Note:

- PART A and PART B: Students are allowed to pick one experiment from PART A and one experiment from PART B, with equal opportunity.
- Change of experiment is allowed only once and marks allotted for procedure to be made zero of the changed part only.
- Marks Distribution:
 - Conduction and Result : Part A-15 Marks, Part B-20 Marks
 - Write-up : 8 Marks
 - Viva :7 Marks